



EUROPEAN
SPALLATION
SOURCE

ESS Data Management & Software Center

- en dansk del af ESS

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NEXMAP, DTU Fysik & ESS DMSC



With input from
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Thomas Holm Rod, ESS
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The Data Management and Software Centre



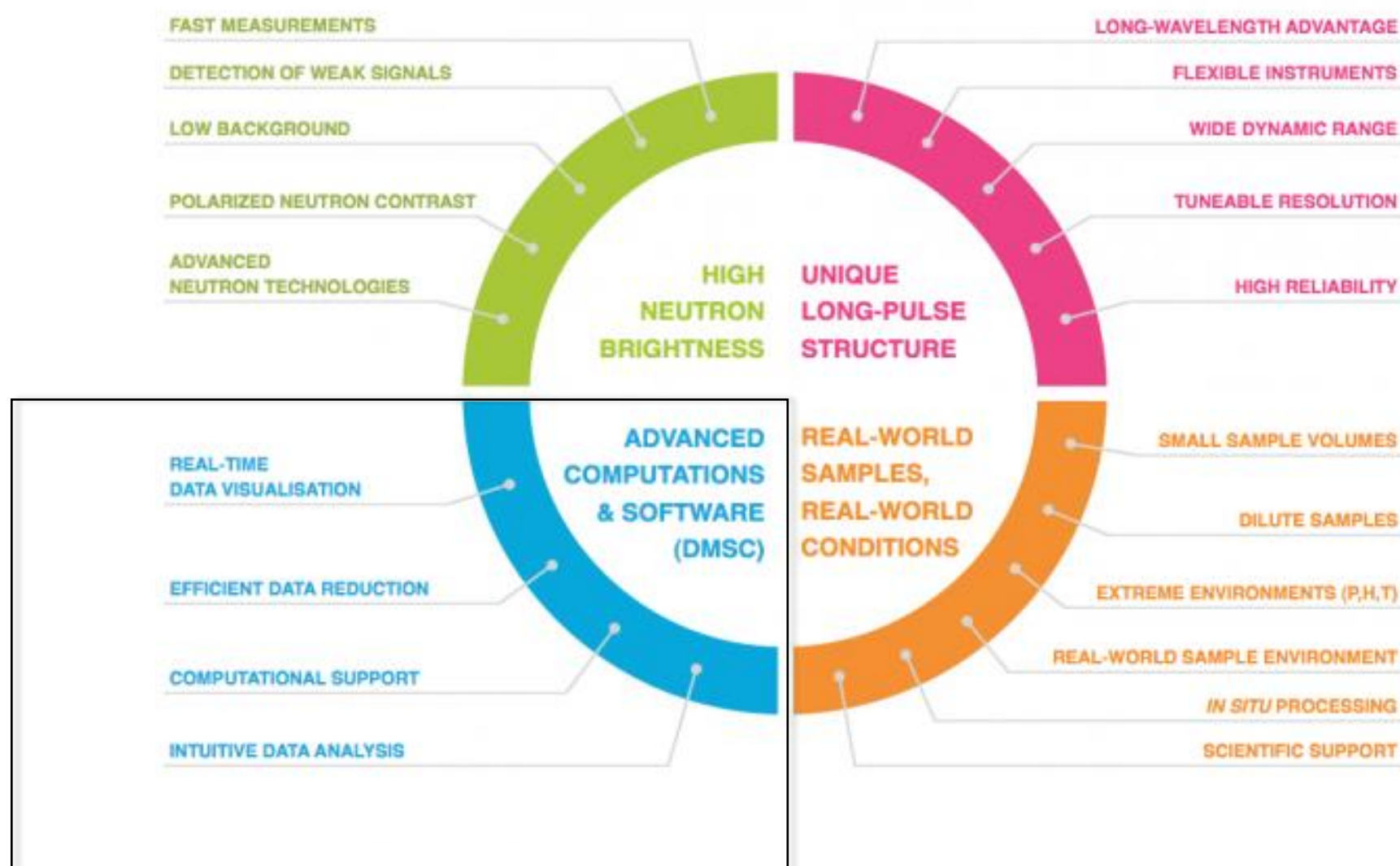
- Data Management
- Instrument Control
- Data Reduction & Analysis
- Instrument simulations
- (Theory and simulations)

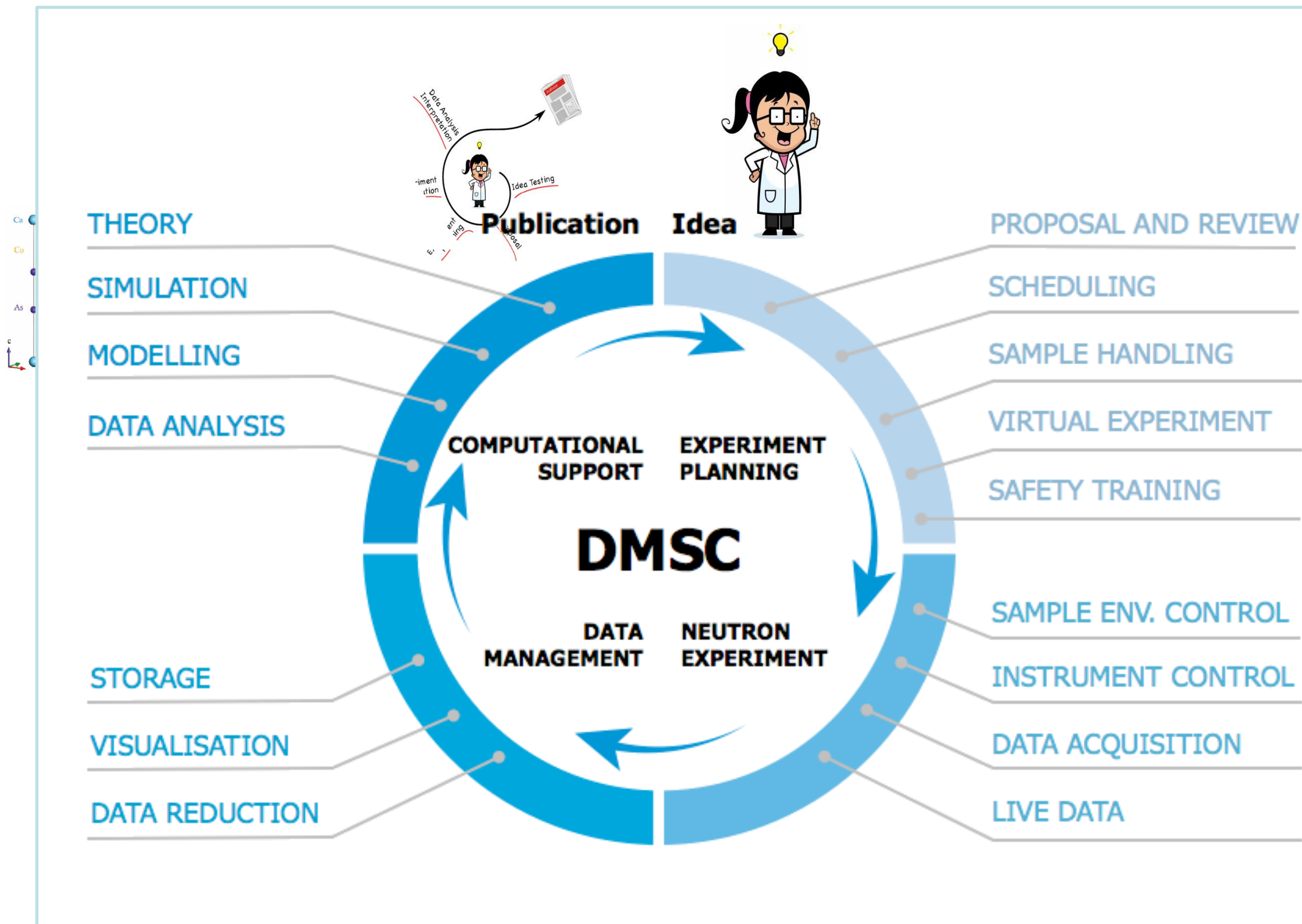


Built on DK competences!

Technical Design Report (2013)

Importance of computing was emphasized already in design phase

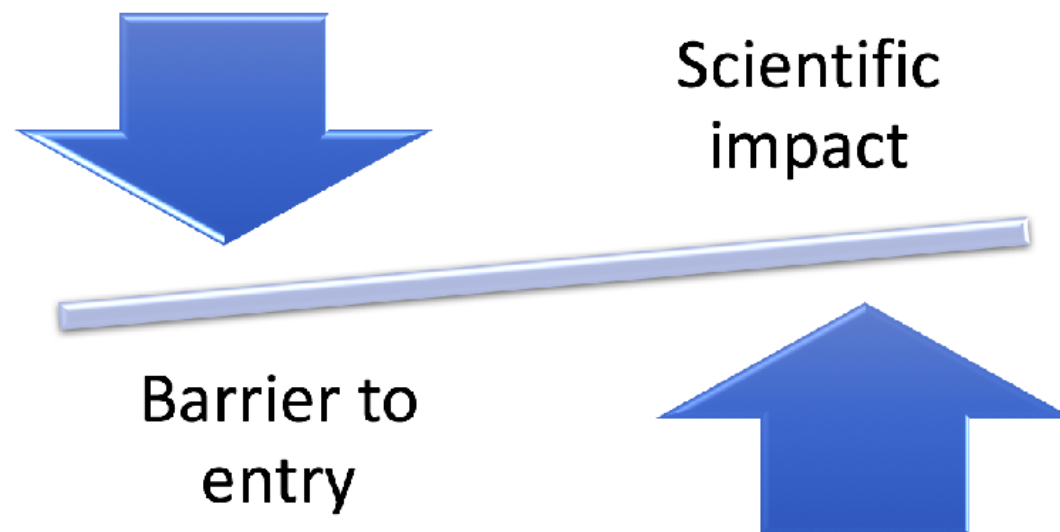




DMSC objective



Minimize the time it takes to analyze and interpret experimental data



This is particular important for neutron sources due to the cost of producing neutrons

Maximise the scientific impact and success of ESS by serving the needs of both non-expert and advanced users

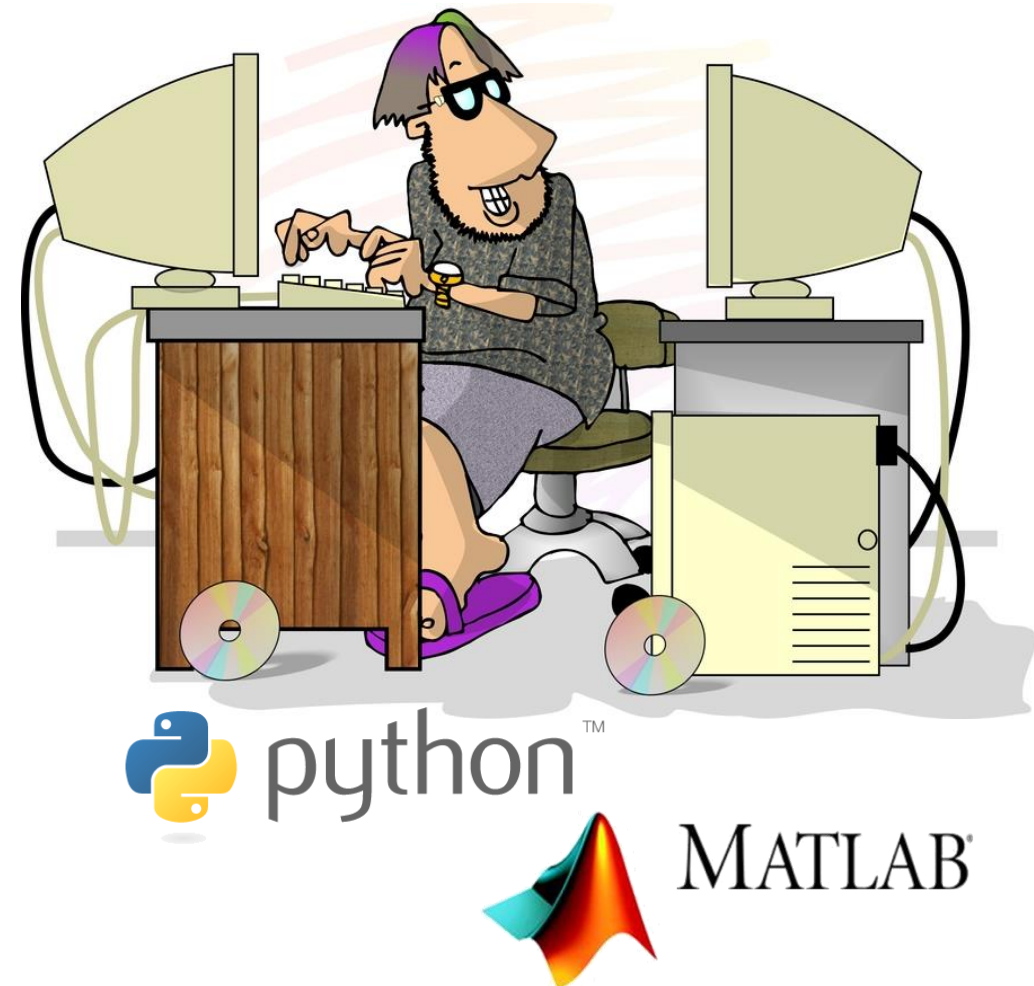
The user community is diverse – even for each instrument

Non-expert and
occasional user (few
days/year)



Graphical User Interface

Experienced (daily) or
geeky user



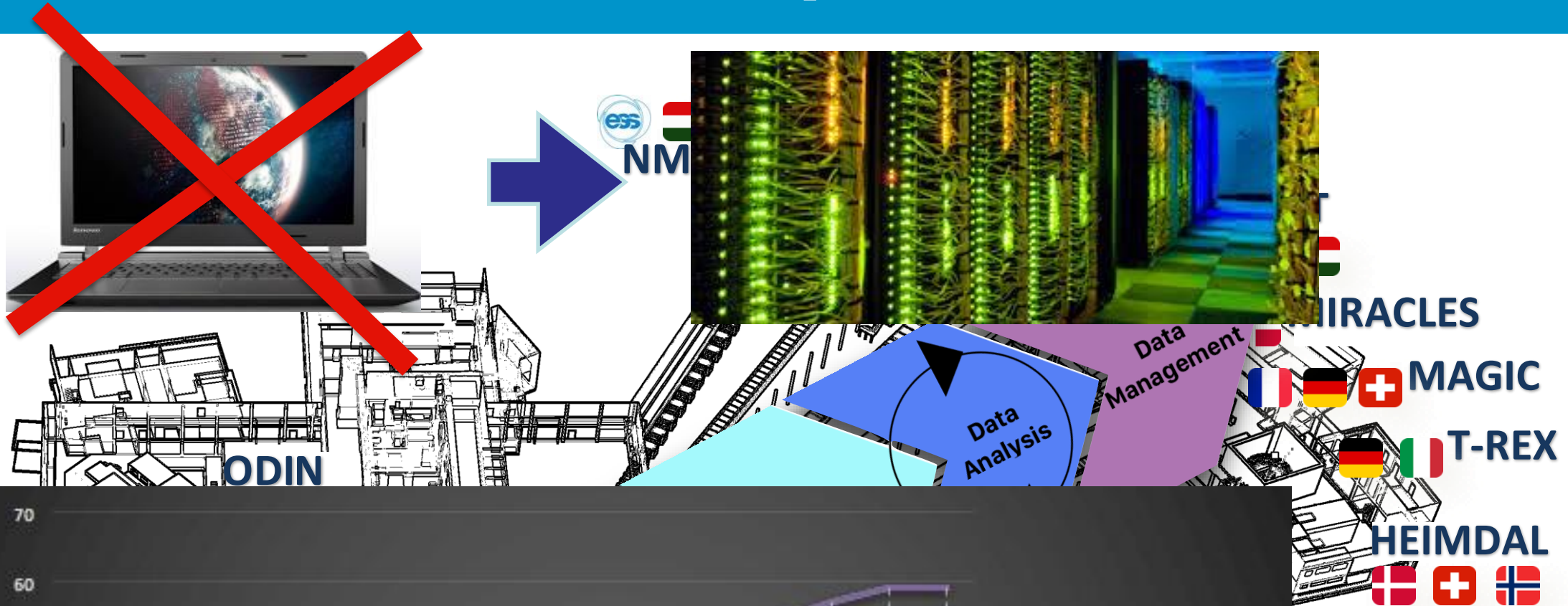
Command Line Interface

DMSC

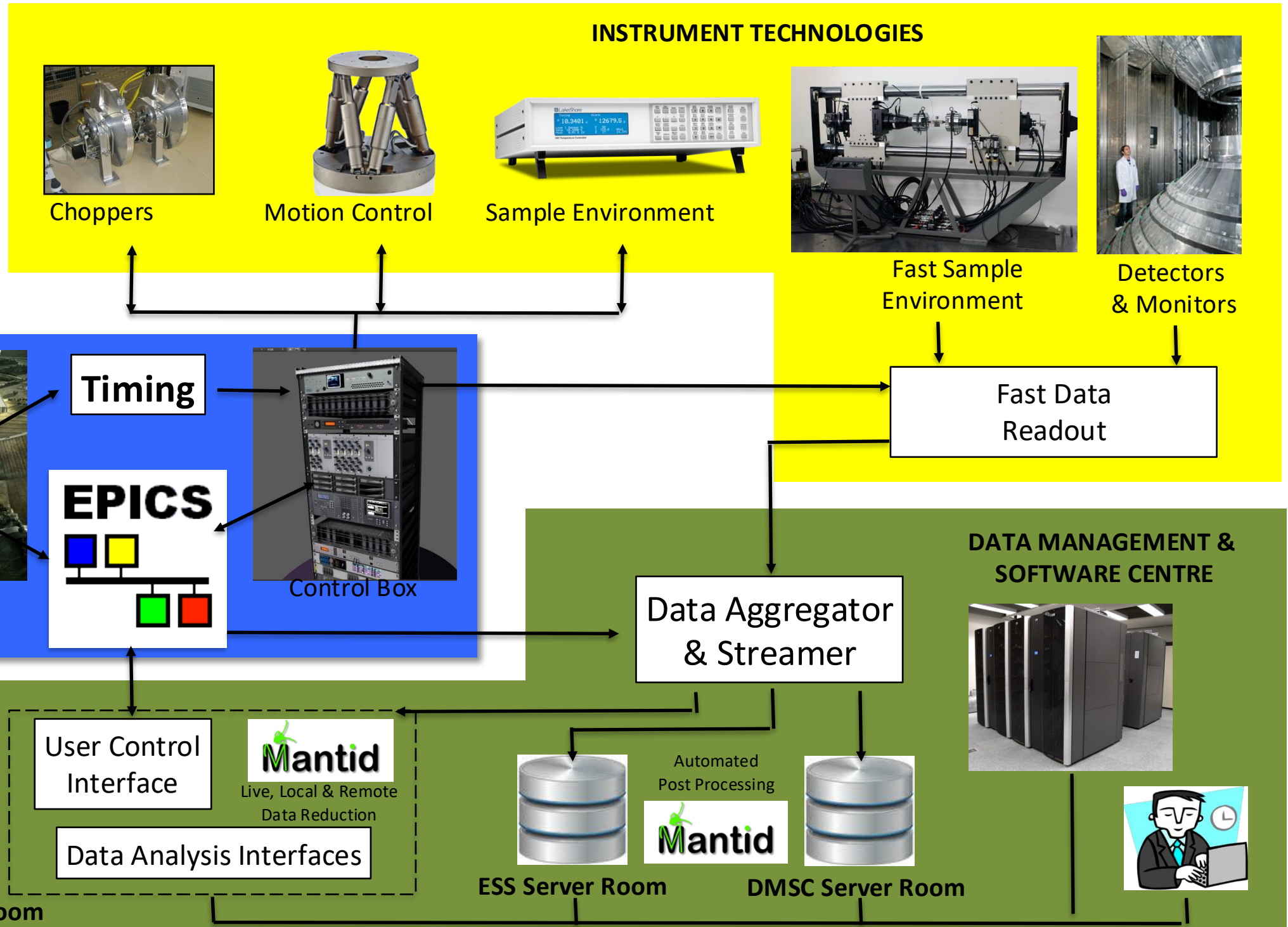
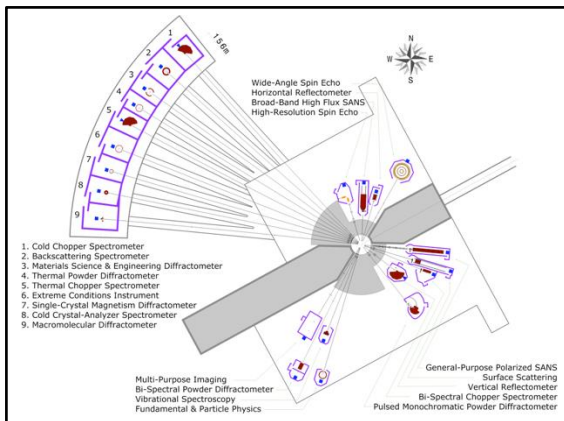
Facility datarates:
~ TB / instrument / day
~ PB / instrument/ year



TO DO:
Scientific
computing
and software
for ALL of this



DMSC's Domain and Interfaces



We must cater for many different scientific domains

Instr.	Tasks	Programs
	ODIN	Im
	BEER	Diffraction
	DREAM	
	HEIMDAL	
	MAGIC	
	NMX	
	LoKI	Large-scale
	SKADI	
	ESTIA	
	FREIA	
	C-SPEC	Spectroscopy
	VOR	
	T-REX	
	BIFROST	
	MIRACL	
	VESPA	

Computed Tomaggrapy, ToF, diffraction imaging, polarization

- 2D Rietveld
- Resolution func.
- Time-of-flight

- MuhRec
- FullProf
- Esmeralda
- Phenix

- 2D Fitting
- Polarization
- Ab initio model

- SASView
- CCP-SAS
- ?

- Resolution func.
- Spin dynamics
- Molecular dyn.
- DFT

- MANTID
- VATES/HORACE
- SpinW
- LAMMPS
- VASP

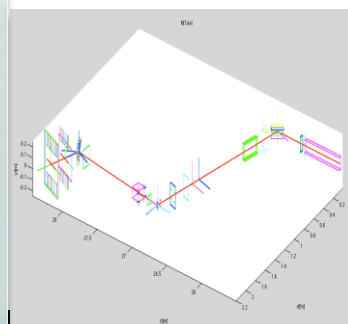
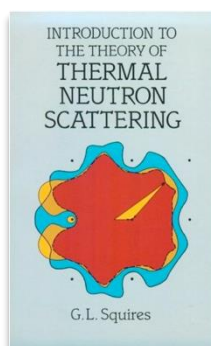
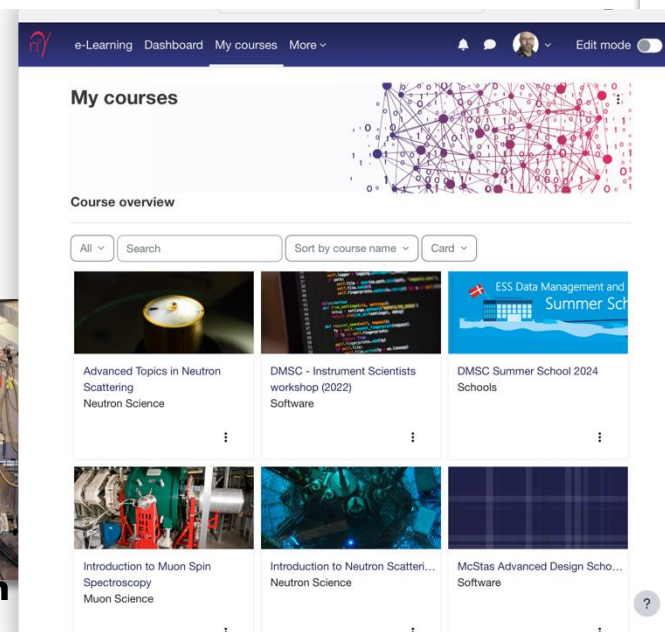
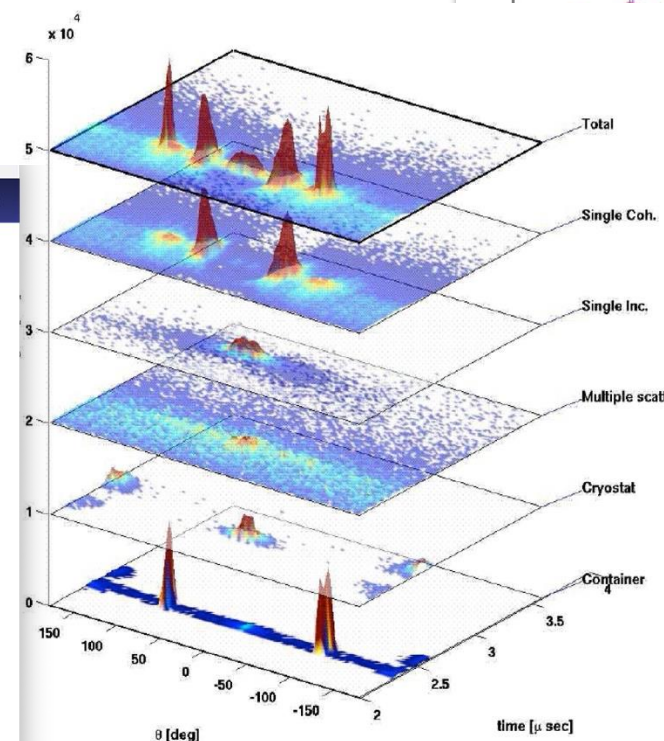
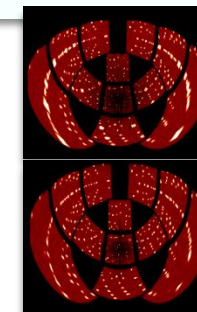
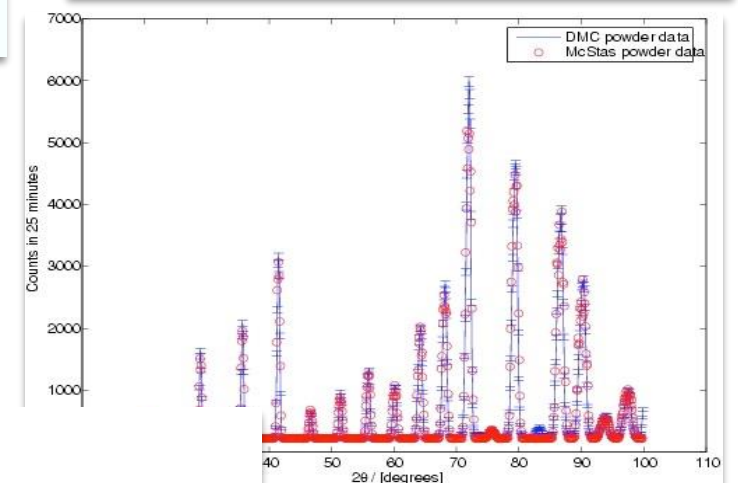
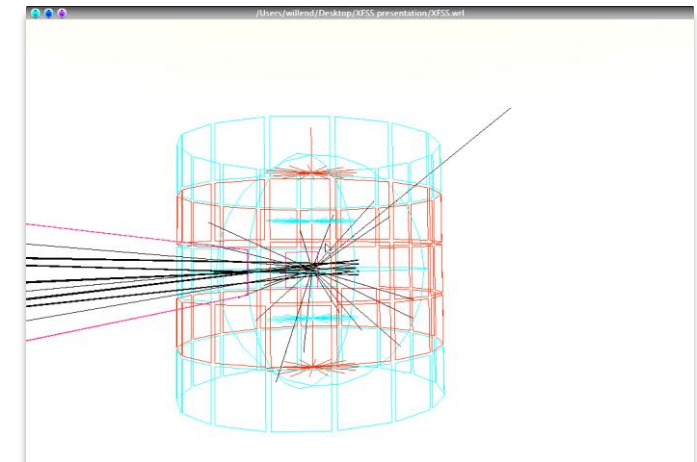
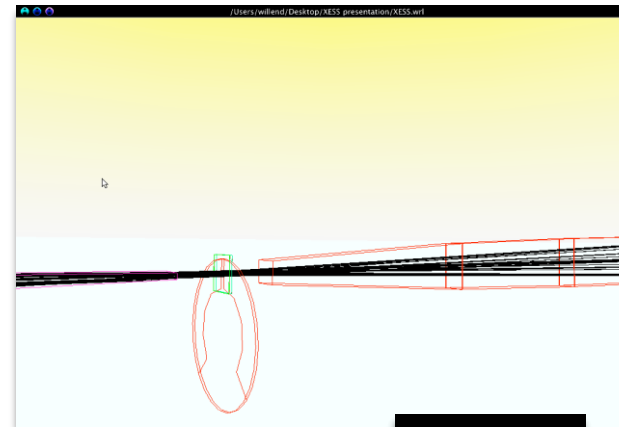
(Incomplete list)

What is McStas used for?



- Instrumentation
- Virtual experiments
- Data analysis
- Teaching

(KU, DTU)



Simulation

Scientific computing gives understanding

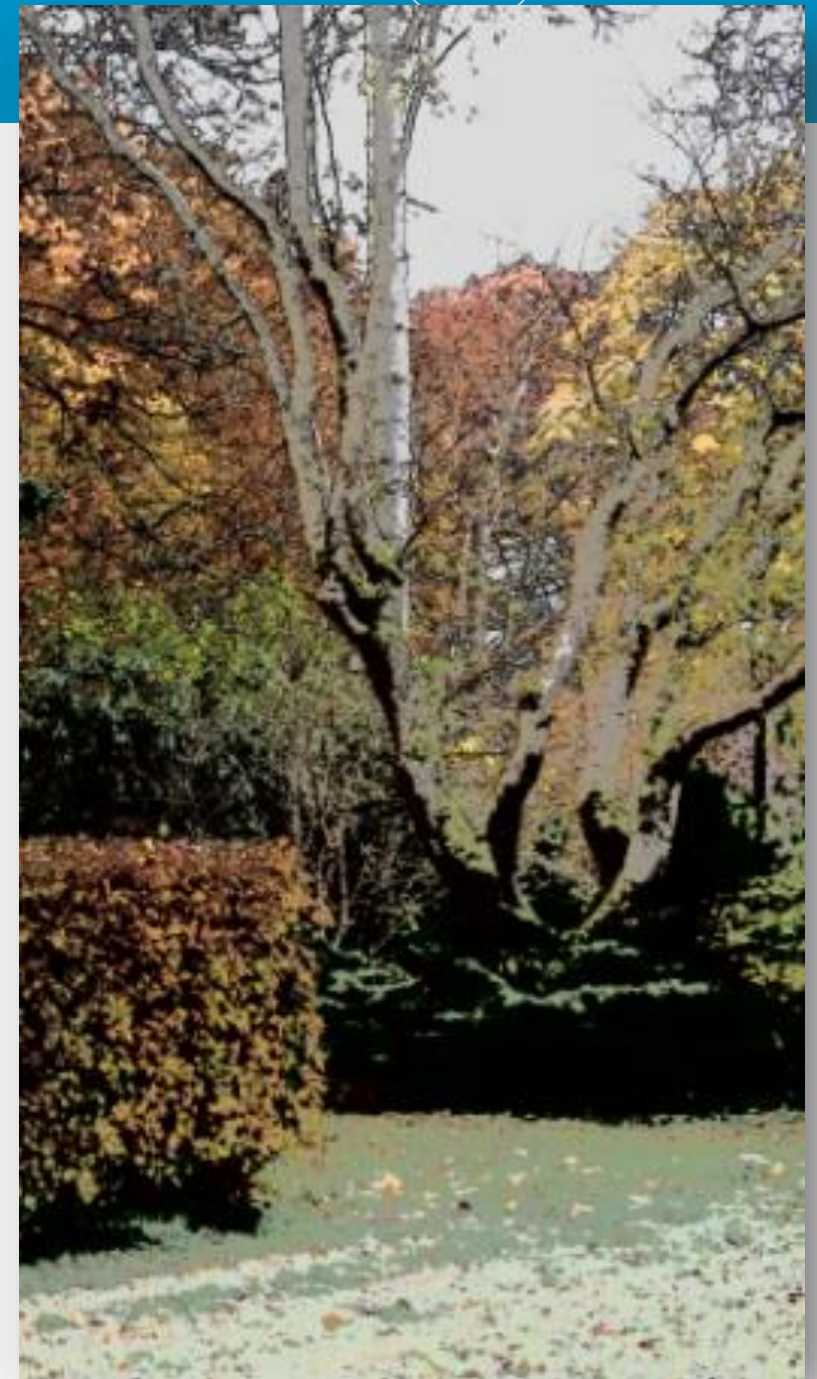


Experiment

An approximate
representation of reality



Reality



Modelling

A cartoon representation of
reality



Questions

